Bin Ren

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Research Interests

Low-level Vision, Scene Understanding, Multimodal Reasoning, Embodied World Modeling.

Education

- **University of Pisa (UNIPI) & University of Trento (UNITN)** *11, 2021 - 10, 2025*
  National Ph.D. in AI for Society Italy
  - Supervised by: Prof. Dr. *Nicu Sebe* and Prof. Dr. *Rita Cucchiara*.
  - Thesis Title: Representation Learning Via Transformer: From 2D to 3D
- **Peking University (PKU)** *09, 2017 - 01. 2021*
  MS in Computer Applied Technology China
  - Supervised by: Prof. Dr. *Hong Liu*.
  - Thesis Title: Grouped Spatial-Temporal Enhancement Modeling Based Human Action Recognition
  - Awards: The 2nd prize of the May 4th Youth Science Award (Challenge Cup of PKU) in 06, 2019
- **Central South University (CSU)** *09, 2012 - 07, 2016*
  BA in Microelectronics Manufacturing Engineering China
  - Scholarship: Three times a 3rd-class scholarship winner in 2013, 2014, and 2015.

Academic Positions

- **École Polytechnique Fédérale de Lausanne (EPFL)** *10, 2026 - now*
  Postdoc Associate (EPFL AI Center Postdoctoral Fellowship) Lausanne, Switzerland
  - Supervised by: Prof. Dr. *Amir Zamir*, Prof. Dr. *Martin Schrimpf*, and Prof. Dr. *Maria Brbić*.
  - Topics: Multi-Modal Learning, Embodied World Modeling
- **Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)** *01, 2026 - 10, 2026*
  Postdoc Associate Abu Dhabi, United Arab Emirates
  - Supervised by: Prof. Dr. *Salman Khan*
  - Topics: VLM(A), Multi-Modal Learning, Embodied World Modeling
- **Institute for Computer Science, Artificial Intelligence and Technology (INSAIT)** *04, 2024 - 09, 2025*
  Visiting Ph.D. Student Sofia, Bulgaria
  - Supervised by: Prof. Dr. *Luc Van Gool*
  - Topics: Low-level Vision, Large Multimodal Models, 3D Representation Learning
- **Eidgenössische Technische Hochschule Zürich (ETH Zürich)** *08, 2023 - 01, 2024*
  Visiting Ph.D. Student Zürich, Switzerland
  - Supervised by: Prof. Dr. *Luc Van Gool*
  - Topics: Low-level Vision, 3D Representation Learning
- **University of Trento (UNITN)** *05, 2021 - 10, 2021*
  Research Assistant (funded by research grant) Trento, Italy
  - Supervised by: Prof. Dr. *Nicu Sebe*
  - Topics: Image Generation, Generative Model

SELECTED PUBLICATIONS

C=CONFERENCE, J=JOURNAL, *=CORRESPONDING AUTHOR

[C.1] **Bin Ren**, Runyi Yang, Qi Ma, Xu Zheng, Mengyuan Liu, Danda Pani Paudel, Luc Van Gool, Rita Cucchiara, Nicu Sebe. **Rethinking Expressivity and Degradation-Awareness in Attention for All-in-One Blind Image Restoration**. Accepted by **ICLR'2026**.

[C.2] **Bin Ren**, Yawei Li, Xu Zheng, Yuqian Fu, Danda Pani Paudel, Hong Liu, Ming-Hsuan Yang, Luc Van Gool, and Nicu Sebe. **Manifold-aware Representation Learning for Degradation-agnostic Image Restoration**. Accepted by **ICLR'2026**.

[C.3] **Bin Ren**, Xiaoshui Huang, Mengyuan Liu, Hong Liu, Fabio Poiesi, Nicu Sebe, and Guofeng Mei. **Masked Clustering Prediction for Unsupervised Point Cloud Pre-training**. **AAAI'2026** Oral.

[C.4] **Bin Ren**, Yawei Li, Jingyun Liang, Rakesh Ranjan, Mengyuan Liu, Rita Cucchiara, Luc Van Gool, Ming-Hsuan Yang, and Nicu Sebe. **Sharing Key Semantics in Transformer Makes Efficient Image Restoration**. **NeurIPS'2024**.

[C.5] **Bin Ren**, Yahui Liu, Yue Song, Wei Bi, Rita Cucchiara, Nicu Sebe, and Wei Wang. **Masked Jigsaw Puzzle: A Versatile Position Embedding for Vision Transformers**. **CVPR'2023**.

[C.6] **Bin Ren**, Guofeng Mei, Danda Pani Paudel, Weijie Wang, Yawei Li, Mengyuan Liu, Rita Cucchiara, Luc Van Gool, and Nicu Sebe. **Bringing masked autoencoders explicit contrastive properties for point cloud self-supervised learning**. **ACCV'2024**.

[C.7] **Bin Ren**, Hao Tang, Yiming Wang, Xia Li, Wei Wang, and Nicu Sebe. **PI-Trans: Parallel-ConvMLP and Implicit-Transformation Based GAN for Cross-View Image Translation**. **ICASSP'2023**.

[C.8] **Bin Ren**, Hao Tang, and Nicu Sebe. **Cascaded Cross MLP-Mixer GANs for Cross-View Image Translation**. **BMVC'2021** Oral.

[C.9] Davide Lobba, Fulvio Sanguigni, **Bin Ren***, Marcella Cornia, Rita Cucchiara, and Nicu Sebe. **Inverse Virtual Try-On: Generating Multi-Category Product-Style Images from Clothed Individuals**. **ICLR'2026**.

[C.10] Davide Lobba, Fulvio Sanguigni, **Bin Ren***, Marcella Cornia, Nicu Sebe, and Rita Cucchiara. **Dress-ED: Instruction Guided Editing for Virtual Try-On and Try-Off**. Accepted by **ECCV'2026** Oral.

[C.11] Yan Shu, **Bin Ren***, Zhitong Xiong, Xiaoxiang Zhu, Begum Demir, Nicu Sebe, and Paolo Rota. **TerraScope: Pixel-Grounded Visual Reasoning for Earth Observation**. Accepted by **CVPR'2026**.

[C.12] Mengjiao Ma, Qi Ma, Yue Li, Jiahuan Cheng, Runyi Yang, **Bin Ren***, Nikola Popovic, Mingqiang Wei, Nicu Sebe, Luc Van Gool, Theo Gevers, Martin R Oswald, Danda Pani Paudel. **SceneSplat++: A Large Dataset and Comprehensive Benchmark for Language Gaussian Splatting**. **NeurIPS'2025**.

[C.13] Yue Li, Qi Ma, Runyi Yang, Huapeng Li, Mengjiao Ma, **Bin Ren***, Nikola Popovic, Nicu Sebe, Ender Konukoglu, Theo Gevers, Luc Van Gool, Martin R Oswald, Danda Pani Paudel. **Scenesplat: Gaussian splatting-based scene understanding with vision-language pretraining**. **ICCV'2025** Oral.

[C.14] Yue Li, Qi Ma, Runyi Yang, Mengjiao Ma, **Bin Ren**, Nikola Popovic, Nicu Sebe, Theo Gevers, Luc Van Gool, Danda Pani Paudel, and Martin R Oswald. **Chorus: Multi-Teacher Pretraining for Holistic 3D Gaussian Scene Encoding**. Accepted by **CVPR'2026** Oral, Award Candidate.

[C.15] Yuqian Fu, Runze Wang, **Bin Ren**, Guolei Sun, Biao Gong, Yanwei Fu, Danda Pani Paudel, Xuanjing Huang, Luc Van Gool **ObjectRelator: Enabling Cross-View Object Relation Understanding Across Ego-Centric and Exo-Centric Perspectives**. **ICCV'2025** Highlight.

[C.16] Qi Ma, Yue Li, **Bin Ren***, Nicu Sebe, Ender Konukoglu, Theo Gevers, Luc Van Gool, and Danda Pani Paudel. **ShapeSplat: A Large-scale Dataset of Gaussian Splats and Their Self-Supervised Pretraining**. **3DV'2025** Oral.

[C.17] Chenfei Liao, Wensong Wang, Zichen Wen, Xu Zheng, Yiyu Wang, Haocong He, Yuanhuiyi Lyu, Lutao Jiang, Xin Zou, Yuqian Fu, **Bin Ren**, Linfeng Zhang, and Xuming Hu **Are We Using the Right Benchmark: An Evaluation Framework for Visual Token Compression Methods**. Accepted by **ACL'2026**.

[C.18] Mengyi Zhao, Mengyuan Liu, **Bin Ren**, Shuling Dai, and Nicu Sebe. **Modiff: Action-conditioned 3d motion generation with denoising diffusion probabilistic models**. **ICASSP'2024**.

[C.19] Chang Liu, Mengyi Zhao, **Bin Ren**, Mengyuan Liu, and Nicu Sebe. **Spatio-Temporal Graph Diffusion for Text-Driven Human Motion Generation**. **BMVC'2023** Oral.

[J.1] **Bin Ren**, Eduard Zamfir, Zongwei Wu, Yawei Li, Yidi Li, Danda Pani Paudel, Radu Timofte, Ming-Hsuan Yang, Luc Van Gool, and Nicu Sebe. **Any Image Restoration via Efficient Spatial-Frequency Degradation Adaptation**. Accepted by **TMLR**, 2026.

[J.2] **Bin Ren**, Hao Tang, Fanyang Meng, Runwei Ding, Philip Torr, and Nicu Sebe. **Cloth interactive transformer for virtual try-on**. **ACM ToMM**, 2023.

[J.3] Weixin Ye, Wei Wang, Yahui Liu, Yue Song, **Bin Ren***, Wei Bi, Rita Cucchiara, and Nicu Sebe. **A Unified Masked Jigsaw Puzzle Framework for Vision and Language Models**. **TPAMI**, 2025.

[J.4] Hao Tang, **Bin Ren**, and Nicu Sebe. **A pure MLP-Mixer-based GAN framework for guided image translation**. **PR**, 2024.

[J.5] Hao Tang, **Bin Ren**, Pingping Wu and Nicu Sebe. **Hierarchical Cross-Attention Network for Virtual Try-On**. **TMM**, 2025.

[J.6] Ti Wang, Mengyuan Liu, Hong Liu, **Bin Ren**, Yingxuan You, Wenhao Li, Nicu Sebe, and Xia Li. **Uncertainty-Aware Testing-Time Optimization for 3D Human Pose Estimation**. **TMM**, 2025.

[J.7] Wenhao Li, Mengyuan Liu, Hong Liu, **Bin Ren**, Xia Li, Yingxuan You, and Nicu Sebe. **HYRE: Hybrid Regressor for 3D Human Pose and Shape Estimation**. **TIP**, 2024.

Academic Services

- Workshop/Challenge Organizer: NTIRE at CVPR'(24,25,26), AIM at ICCV'25, SPI (1st) at SWC'24.
- Conference Reviewer: ICLR, ICML, NeurIPS, CVPR, ICCV, ECCV, ACL, ACCV, BMVC, 3DV, and ICASSP, etc.
- Journal Reviewer: TPAMI, TIP, IJCV, RAL, TMM, and ToMM, etc.
- Area Chair: BMVC'25

Skills

- Programming: Python, PyTorch, Linux, Huggingface, Git, ViM, ROS2.
- ML/AI: ViT, Low-Level Vision, 3D Vision, Self-Supervised Learning, Multimodal Reasoning, World Model.
- Teaching: Advanced Computer Vision (Teaching Assistant, Winter, 2022, UNITN)

Additional Information

Languages: English (C1, Frequent use), Italian (A2, Basic), Mandarin (Native).
Interests: Cooking, Hiking, and Singing.